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Electronic supplementary
material from: Theoretical
insights into the electronic
structure and NBO features
driving nonlinear optical
enhancement in the
donor– π –acceptor system of
OPPh₃py

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Table S1. Optimized bond lengths (Å) for OPPh₃py at the DFT B3LYP-D3(BJ)/6-311++G(2df,p) level of theory.

Bond (Å)	Experimental		Optimized Conformation
	Isomer I	Isomer II	
P1–O2	1.487 (17)	1.478 (18)	1.510
P1–C3	1.798 (2)	1.805 (3)	1.827
P1–C14	1.798 (3)	1.797 (3)	1.826
P1–C25	1.808 (2)	1.806 (3)	1.832
C3–C4	1.364 (3)	1.381 (4)	1.399
C3–C12	1.378 (4)	1.380 (4)	1.399
C4–C6	1.382 (4)	1.386 (4)	1.394
C6–C8	1.365 (4)	1.358 (4)	1.395
C8–C10	1.338 (4)	1.361 (5)	1.395
C10–C12	1.374 (4)	1.378 (5)	1.393
C14–C23	1.387 (4)	1.382 (4)	1.399
C14–C15	1.391 (3)	1.381 (3)	1.399
C15–C17	1.370 (4)	1.382 (4)	1.393
C17–C19	1.372 (5)	1.363 (4)	1.394
C19–C21	1.364 (4)	1.365 (4)	1.395
C21–C23	1.385 (4)	1.379 (4)	1.393
C25–C33	1.391 (3)	1.387 (3)	1.400
C25–C26	1.416 (3)	1.398 (3)	1.414
C26–C27	1.386 (4)	1.382 (4)	1.397
C26–C35	1.516 (4)	1.522 (4)	1.524
C27–C29	1.374 (4)	1.381 (4)	1.394
C29–C31	1.364 (4)	1.360 (4)	1.391
C31–C33	1.376 (3)	1.380 (4)	1.393
C35–N38	1.431 (3)	1.442 (3)	1.457
N38–C40	1.350 (4)	1.357 (3)	1.373
C40–N41	1.335 (3)	1.326 (3)	1.342
C40–C48	1.391 (4)	1.398 (4)	1.416
N41–C42	1.339 (4)	1.350 (4)	1.339
C42–C44	1.346 (5)	1.350 (5)	1.387
C44–C46	1.366 (6)	1.370 (5)	1.402
C46–C48	1.363 (5)	1.348 (4)	1.380

^a Numbers in parentheses represent estimated standard uncertainties in the last digit(s) of the value.

Table S2. Angles (°) for OPPh₃py at the DFT B3LYP-D3(BJ)/6-311++G(2df,p) level of theory.

Angle (°)	Experimental		Optimized Conformation
	Isomer I	Isomer II	
O2-P1-C3	112.390 (110)	112.620 (120)	112.331
O2-P1-C14	110.160 (110)	111.44 (120)	111.294
C3-P1-C14	106.340 (120)	106.420 (120)	106.039
O2-P1-C25	112.550 (110)	114.120 (110)	114.132
C3-P1-C25	106.980 (110)	104.94 (110)	105.433
C14-P1-C25	108.140 (120)	106.730 (120)	107.069
C4-C3-C12	117.700 (200)	118.000 (300)	119.881
C4-C3-P1	124.100 (200)	118.300 (200)	122.511
C12-C3-P1	118.000 (200)	123.600 (200)	117.608
C3-C4-C6	120.900 (200)	120.500 (300)	119.973
C8-C6-C4	119.800 (300)	120.500 (300)	120.023
C10-C8-C6	120.200 (300)	119.700 (300)	120.091
C8-C10-C12	120.200 (300)	120.500 (300)	120.093
C10-C12-C3	121.200 (300)	120.800 (300)	119.936
C23-C14-C15	118.800 (300)	118.400 (300)	119.829
C23-C14-P1	123.300 (200)	119.400 (200)	123.083
C15-C14-P1	117.900 (200)	122.300 (200)	117.089
C17-C15-C14	120.400 (300)	120.300 (300)	119.968
C15-C17-C19	120.300 (300)	120.200 (300)	120.089
C21-C19-C17	120.200 (300)	120.500 (300)	120.059
C19-C21-C23	120.300 (300)	119.400 (300)	120.052
C21-C23-C14	120.000 (300)	121.200 (300)	119.998
C33-C25-C26	118.800 (200)	119.200 (200)	119.850
C33-C25-P1	120.600 (200)	120.300 (190)	120.190
C26-C25-P1	120.590 (190)	120.450 (190)	119.958
C27-C26-C25	118.000 (200)	118.200 (200)	118.458
C27-C26-C35	120.200 (200)	121.400 (200)	118.822
C25-C26-C35	121.700 (200)	120.400 (200)	122.663
C29-C27-C26	121.500 (300)	121.800 (300)	121.350
C31-C29-C27	120.800 (300)	120.000 (300)	120.013
C29-C31-C33	119.100 (300)	119.400 (300)	119.596
C31-C33-C25	121.700 (300)	121.400 (300)	120.732
N38-C35-C26	115.500 (200)	115.400 (200)	113.203
C40-N38-C35	122.200 (300)	122.900 (200)	121.748
N41-C40-N38	117.700 (300)	118.100 (300)	118.287
N41-C40-C48	121.600 (300)	122.900 (300)	121.988
N38-C40-C48	120.700 (300)	119.100 (200)	119.706
C40-N41-C42	117.300 (300)	115.900 (300)	117.994
N41-C42-C44	124.800 (400)	124.600 (300)	124.338
C42-C44-C46	117.400 (400)	118.400 (300)	117.398
C48-C46-C44	120.600 (400)	119.500 (300)	119.574
C46-C48-C40	118.400 (400)	118.800 (300)	118.702

^a Numbers in parentheses represent estimated standard uncertainties in the last digit(s) of the value.

The following molecules (Table S3) were selected to benchmark the global reactivity parameters of OPPh₃py. These molecules were chosen because of their similarity in structural formula, each having triphenylphosphine oxide as the base structural unit. Although the reference molecules selected share a common structural framework with OPPh₃py, their reported reactivity parameters were not obtained at the same level of theory used in our calculations (B3LYP-D3(BJ)/6-311G++(2df,p)). As a result, the comparison presented here should be regarded as qualitative rather than quantitative. Nonetheless, structural similarity among the chosen molecules provides a reasonable basis for establishing general trends. To strengthen the validity of our analysis, all comparisons are interpreted cautiously, and the limitations arising from cross-method data are explicitly acknowledged.

Table S3. Molecular structures of the reference compounds used for benchmarking the global reactivity parameters of OPPh₃py.

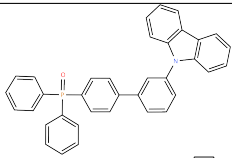
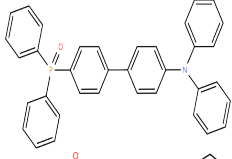
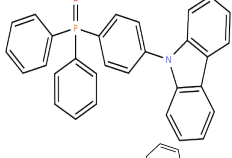
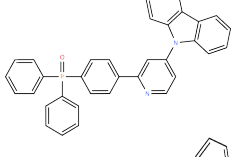
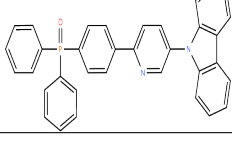
Molecule	Rename	Level of Theory	Structural Formula
1 Ref. [1]	TPP1	B3LYP/ 6-31G(d,p)	
POA Ref. [2]	TPP2	B3LYP/ 6-31G(d)	
MPO12 Ref. [3]	TPP3	B3LYP/ 6-31G(d)	
pPO-ppy-mCZ Ref. [4]	TPP4	B3LYP/ 6-31G(d)	
pPO-ppy-pCZ Ref. [4]	TPP5	B3LYP/ 6-31G(d)	

Table S4. Local reactivity parameters: Fukui functions using CM5 population analysis: local softness ($s_k^\pm$) and philicity ($\omega_k^\pm$) in eV for selected atomic sites of OPPh₃py.

Atom	Fukui function		Local softness		Philicity	
	f_k^+	f_k^-	s_k^+	s_k^-	ω_k^+	ω_k^-
P1	0.0364	0.0009	0.0104	0.0003	0.0671	0.0017
O2	0.0237	0.0001	0.0068	0.0000	0.0437	0.0002
C3	0.0253	-0.0070	0.0072	-0.0020	0.0467	-0.0129
C4	0.0308	-0.0013	0.0088	-0.0004	0.0568	-0.0024
C6	0.0271	0.0090	0.0078	0.0026	0.0500	0.0166
C8	0.0558	0.0146	0.0160	0.0042	0.1029	0.0269
C10	0.0400	0.0098	0.0115	0.0028	0.0738	0.0180
C12	0.0228	-0.0007	0.0065	-0.0002	0.0420	-0.0013
C14	0.0261	-0.0054	0.0075	-0.0015	0.0481	-0.0099
C15	0.0360	-0.0018	0.0103	-0.0005	0.0664	-0.0032
C17	0.0308	0.0083	0.0088	0.0024	0.0568	0.0153
C19	0.0574	0.0141	0.0164	0.0040	0.1059	0.0260
C21	0.0364	0.0106	0.0104	0.0030	0.0671	0.0196
C23	0.0187	0.0021	0.0054	0.0006	0.0345	0.0039
C25	0.0173	0.0054	0.0050	0.0015	0.0319	0.0099
C26	0.0161	-0.0066	0.0046	-0.0019	0.0297	-0.0122
C27	0.0213	0.0033	0.0061	0.0009	0.0393	0.0061
C29	0.0427	0.0155	0.0122	0.0044	0.0787	0.0286
C31	0.0247	0.0276	0.0071	0.0079	0.0456	0.0509
C33	0.0139	0.0172	0.0040	0.0049	0.0256	0.0317
C35	0.0045	0.0216	0.0013	0.0062	0.0083	0.0398
N38	-0.0017	0.1534	-0.0005	0.0439	-0.0031	0.2829
C40	-0.0006	0.0334	-0.0002	0.0096	-0.0011	0.0616
N41	0.0023	0.0575	0.0007	0.0165	0.0043	0.1060
C42	0.0128	0.0609	0.0037	0.0174	0.0236	0.1123
C44	0.0178	0.1046	0.0051	0.0299	0.0329	0.1929
C46	0.0178	0.0453	0.0051	0.0130	0.0329	0.0836
C48	0.0064	0.0776	0.0018	0.0222	0.0118	0.1432

Table S5. Second order perturbation theory analysis of the Fock matrix in the NBO basis for OPPh₃py.

Donor (<i>i</i>)	Type	ED(<i>i</i>) (e)	Acceptor (<i>j</i>)	ED(<i>j</i>) (e)	Type	^a <i>E</i> (2) (kcal/mol)	^b <i>E</i> (<i>j</i>) − <i>E</i> (<i>i</i>) (a.u)	^c <i>F</i> (<i>i</i> , <i>j</i>) (a.u)
LP O2	-	1.97598	BD* N38-H39	0.02798	σ*	2.94	1.03	0.049
LP O2	-	1.97598	RY P1	0.00939	-	7.03	1.76	0.099
LP O2	-	1.83260	BD* P1-C14	0.11762	π*	17.75	0.5	0.084
LP O2	-	1.83260	RY P1	0.02852	-	8.36	1.26	0.092
LP O2	-	1.83217	BD* P1-C3	0.12524	π*	17.96	0.49	0.084
LP O2	-	1.83217	BD* P1-C25	0.12301	π*	9.49	0.49	0.061
LP O2	-	1.83217	RY* P1	0.02994	-	8.54	1.23	0.091
LP N38	-	1.74306	BD* C26-C35	0.03465	π*	8.2	0.64	0.065
LP N38	-	1.74306	BD* C40-N41	0.46016	π*	53.84	0.26	0.105
LP N41	-	1.90895	BD* C40-C48	0.03238	π*	9.68	0.83	0.08
LP N41	-	1.90895	BD* C42-C44	0.2432	π*	8.53	0.87	0.077
BD C3-C4	π	1.68089	BD* C6-C8	0.32240	π*	18.13	0.29	0.065
BD C3-C12	π	1.68089	BD* C10-C12	0.30291	π*	20.33	0.29	0.069
BD C6-C8	π	1.64999	BD* C3-C4	0.37680	π*	23.02	0.27	0.071
BD C6-C8	π	1.64999	BD* C10-C12	0.30291	π*	19.32	0.28	0.066
BD C10-C12	π	1.64174	BD* C3-C4	0.37680	π*	20.34	0.27	0.066
BD C10-C12	π	1.64174	BD* C6-C8	0.32240	π*	21.79	0.28	0.069
BD C14-C15	π	1.67057	BD* C17-C19	0.31897	π*	17.98	0.29	0.064
BD C14-C15	π	1.67057	BD* C21-C23	0.31491	π*	21.37	0.28	0.07
BD C17-C19	π	1.65047	BD* C14-C15	0.35991	π*	23.08	0.28	0.071
BD C17-C19	π	1.65047	BD* C21-C23	0.31491	π*	19.85	0.28	0.066
BD C21-C23	π	1.65726	BD* C14-C15	0.35991	π*	19.27	0.28	0.065
BD C21-C23	π	1.65726	BD* C17-C19	0.31897	π*	20.91	0.28	0.069
BD C25-C33	π	1.69371	BD* C26-C27	0.30476	π*	19.88	0.3	0.069
BD C25-C33	π	1.69371	BD* C29-C31	0.32040	π*	17.25	0.29	0.063
BD C26-C27	π	1.62887	BD* C25-C33	0.37272	π*	20.03	0.26	0.065
BD C26-C27	π	1.62887	BD* C29-C31	0.32040	π*	22.31	0.27	0.07
BD C29-C31	π	1.65596	BD* C25-C33	0.37272	π*	23.6	0.27	0.072
BD C29-C31	π	1.65596	BD* C26-C27	0.30476	π*	18.8	0.29	0.066
BD C40-N41	π	1.71010	BD* C42-C44	0.31143	π*	30.15	0.32	0.088
BD C42-C44	π	1.69737	BD* C40-N41	0.46016	π*	13.34	0.26	0.053
BD C42-C44	π	1.69737	BD* C46-C48	0.28765	π*	24.22	0.28	0.074
BD C46-C48	π	1.73246	BD* C40-N41	0.46016	π*	28.86	0.27	0.079
BD C46-C48	π	1.73246	BD* C42-C44	0.31143	π*	14.35	0.29	0.057

^a *E*(2) means energy of hyper conjugative interactions^b Energy difference between donor and acceptor *i* and *j* NBO orbitals^c *F*(*i*, *j*) is the Fock matrix element between *i* and *j* NBO orbitals

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